



creo®

How to CAD Almost Anything!

MIT Summer '26

Instructor: Andy Eskenazi

Session 1



AEROASTRO

MIT

Agenda

- Quick intros!
 - Why CAD?
- Workshop logistics and important info
 - Overview of sessions and projects.
- Session 1:
 - Demo of the session's Creo commands.
 - Demo of projects: a toy Xylophone, a Pan flute, a Gong!
- Questions?
- Preview of Session 2



Quick Intros!

The project that inspired the current musical theme!



Me (Andy)

You (in IAP 2026)!

- 1) Name, major, year
 - Eg: Andy, AeroAstro, PhD Y4.
- 2) Hometown
 - Eg: Buenos Aires, Argentina.
- 3) Spirit animal
 - Eg: Angus Aberdeen cow.
- 4) Why CAD almost anything?
 - Eg: I think it's cool to look at a random object and imagine how to cad it!

Why CAD?

“I think it'd be a really cool skill to have, and I'm very interested in learning more about modeling”

“Designing musical instruments seems so cool!”

“will be cool to reverse engineer and customize musical instruments!”

“Learning how to CAD more advanced things”

“I am hoping to learn and be more comfortable with how to use CAD software!”

“Learning Creo, as it is industry standard for me career goals”

“I want to learn how to CAD better, and translate my CAD skills to a new software”

“Mens et Manus”

Why Creo?

- Add another CAD software to the “How to CAD” OCW series.
- Used widely in industry, with lots of functionalities.
- Free for students and educators!

The screenshot shows the MIT OpenCourseWare website. At the top is a black navigation bar with the MIT logo, 'OpenCourseWare', a search icon, a 'GIVE NOW' button, and links for 'ABOUT OCW', 'HELP & FAQs', and 'CONTACT US'. Below this is a blue header bar with the course title 'How To CAD Almost Anything' and the text 'RES.16-002 | January IAP 2024 | Non-Credit'. The main content area is divided into three columns. The left column is a sidebar with a list of CAD software options: SolidWorks, Fusion 360, Inventor, Onshape, Siemens NX, and Instructor Insights, each with a dropdown arrow. The middle column contains the course details: 'Course Description' (a paragraph about designing everyday objects), 'Course Info' (listing the instructor Andy G. Eskenazi and departments Aeronautics and Astronautics, Aerospace Engineering, and Mechanical Engineering), and 'Learning Resource Types' (with buttons for Instructor Insights, Lecture Notes, and Lecture Videos). The right column features a collage of 3D models and a 'Feedback' button. A small text box on the right side of the collage states: 'Students in this workshop learned basic CAD skills and reverse-engineering of an object into a 3D model. (Image courtesy of the instructor; used with permission.)'

- Explore the differences with other CAD software, namely Solidworks!



Welcome!

How to CAD

Almost Anything!

MIT Summer '26



Workshop logistics

Session 1: June 3		Session 2: June 10		Session 3: June 17	
Sketches, basic feature commands, editing and defining sketches, coloring parts, changing material properties	▪ Xylophone ▪ Pan Flute ▪ Gong	Spline tool, sketch picture, sketch projections	▪ Ukelele ▪ Electric Guitar ▪ Morin Khuur	Revising commands from Session 1 & 2	▪ Harmonica ▪ Traditional Violin ▪ Pipa
Session 4: June 24		July 1 - No class!		Session 5: July 8	
Mirroring, circular and rectangular patterns, plane creation	▪ Maracas ▪ 4-string Banjo ▪ Bongos			Loft, sweep	▪ Natural Horn ▪ Trombone
Session 6: July 15		Session 7: July 22		Session 8: July 29	
Revising commands from Session 3 & 4	▪ Accordion ▪ 26-string Harp	Assemblies, exploded view, engineering drawings	▪ Cymbal stand ▪ Bell kit with stands	Sketch/feature names, equations, design table, configurations	▪ Vertical Piano ▪ Euphonium

Important info

- Meeting times:
 - Wednesdays: 3pm – 5pm.
- GIS & Data Lab (Rotch Library)
 - Open M – F: 10:00am – 6:00pm
 - 16 computers (all of which have Inventor on them).
- Online resources:
 - GitHub
 - <https://github.com/andyeske/How-to-CAD-Creo>

How to CAD Almost Anything!

Summer 2026 – AeroAstro Workshop

A compressed yet rewarding introduction to the parametric design software [PTC Creo](#), for beginners (no experience at all) and pro-users alike. Come learn how to CAD (computer-aided design) essentially almost anything!



Yes, this could be YOU at the end of the workshop! You'll be equipped with the tools to design cool looking objects such as an iPod classic, an X-wing fighter, a natural trumpet, a Playmobil figure, a LEGO Tower Bridge, a PlayStation 1 controller, a GameBoy, a Pokémon card, and even a Diet Coke can! These are all projects from "[How to CAD Almost Anything! – IAP 2026](#)".

Workshop Details

Subject Title:	How to CAD Almost Anything!
Prerequisites:	Willingness to have fun and think outside the box!
Enrollment:	20.
Attendance:	Participants must attend all sessions.
Meeting Rooms:	GIS & Data Lab (first floor of the Rotch Library, 7-238).
Meeting Times:	Wednesdays 3pm – 5pm. There is a total of 8 weekly sessions, starting on 06/03/26 until 07/29/26. The sessions will take place on 06/03, 06/10, 06/17, 06/24, 07/08, 07/15, 07/22, and 07/29.

Check out the syllabus if you have more questions!

Session 1



A Pan flute

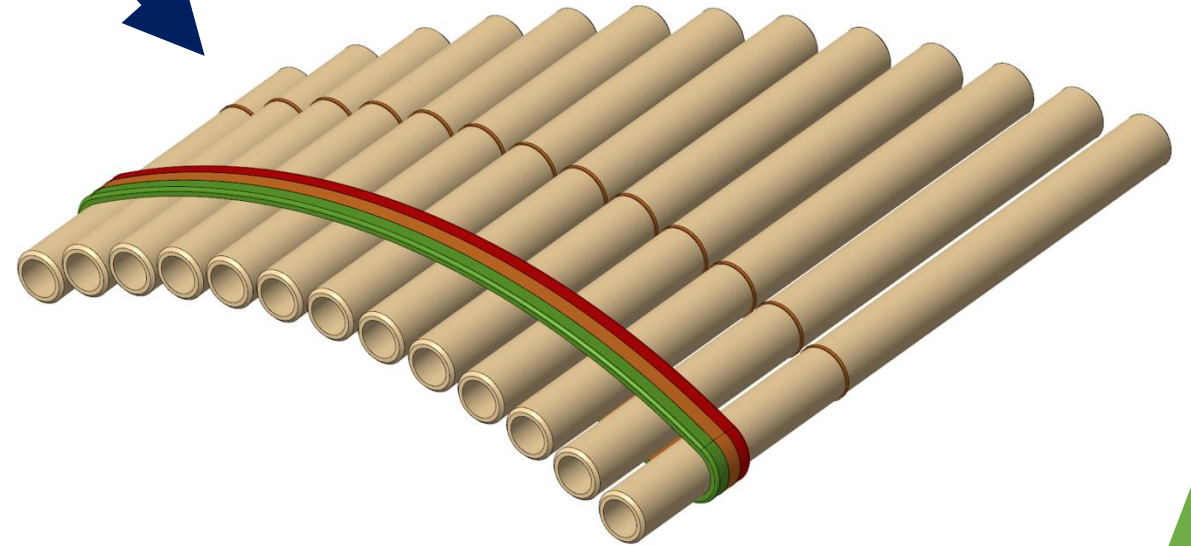
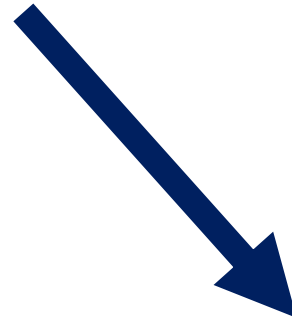


A toy Xylophone

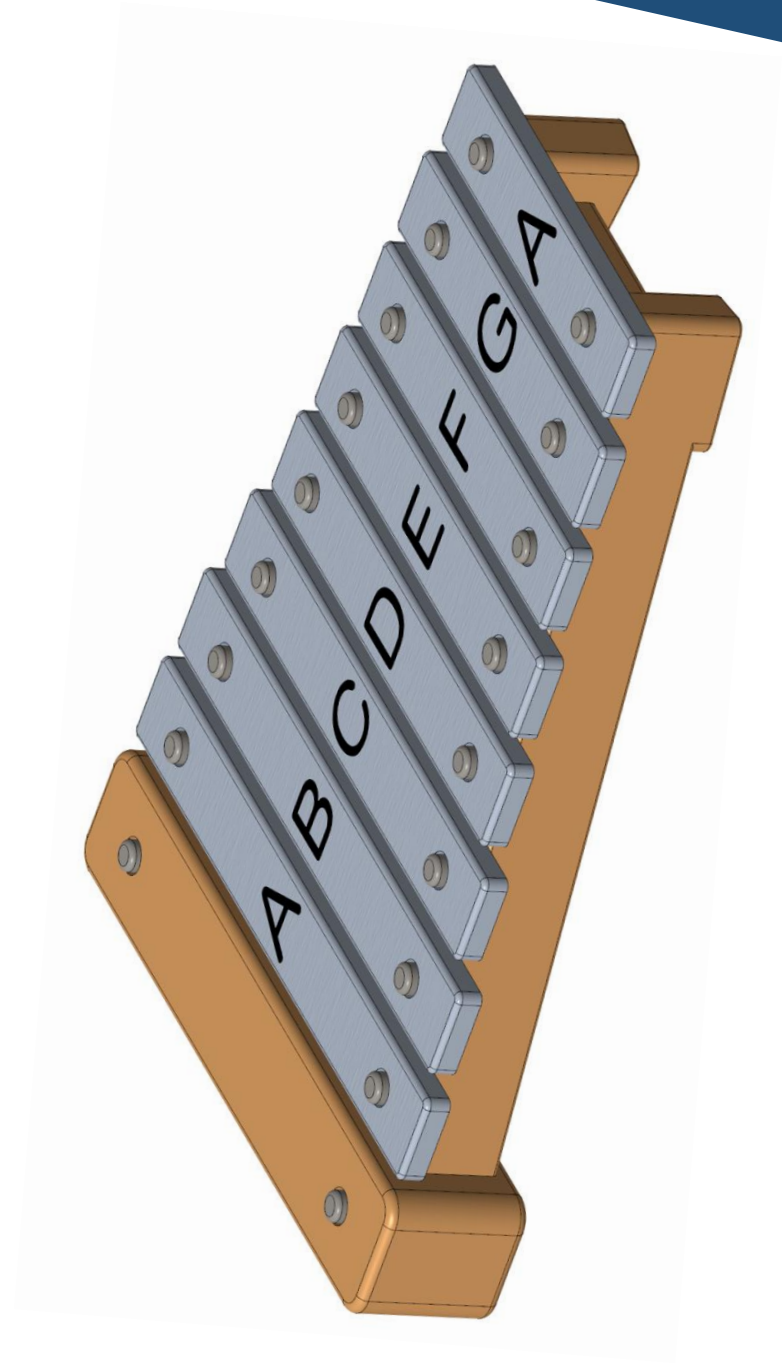


A Gong

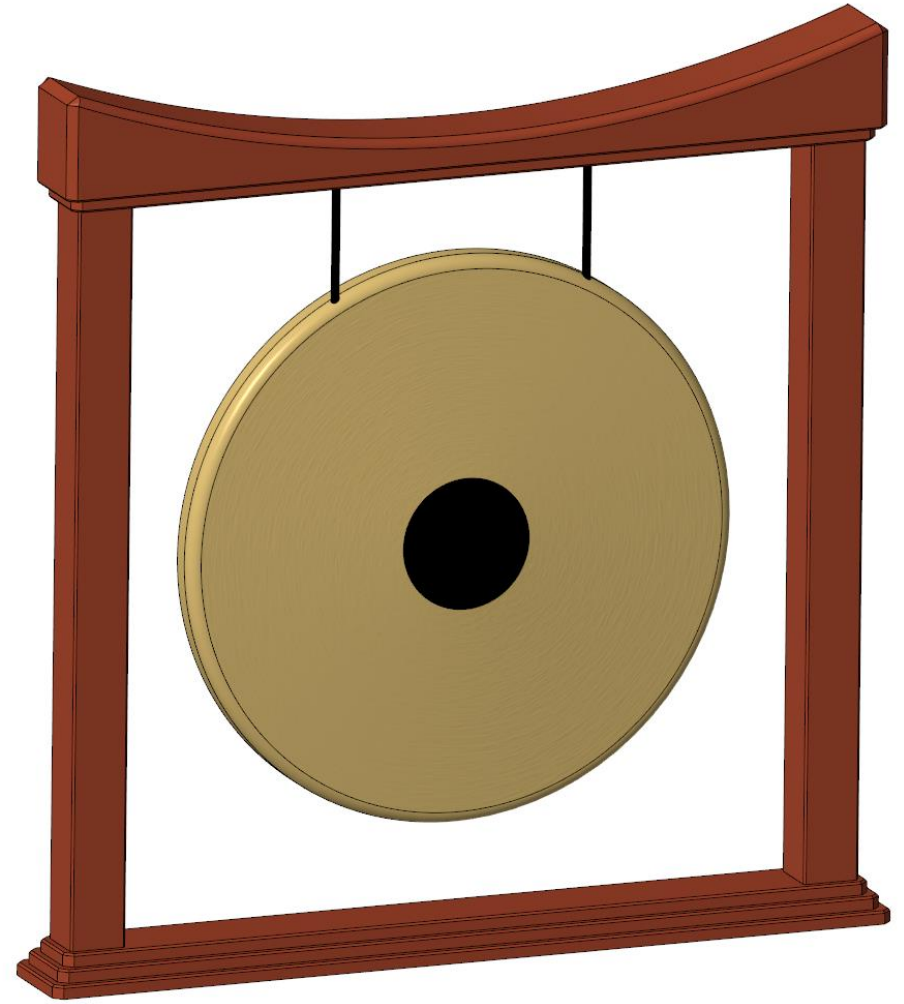
Pan Flute



Toy Xylophone



Gong





Questions?

Preview into Session 2



Electric
Guitar



Ukelele



Morin Khuur